

MARS Standard Communications (MSC) Training Course

February 2021

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In this course, you will:

Demonstrate performing the Mission Essential Tasks (MET) relating to the configuration and operation of the MSC Software.

This training covers installing/updating the MSC software and performing basic operational tasks with the various software components. These tasks are expected of all MARS members in order to support the MARS mission. Performing these tasks will standardize our operations across the enterprise and facilitate leader/trainer problem solving at an enterprise level using common nomenclature and operational concepts.

In most cases, MARS members may have already updated their MSC software to the Feb 1, 2021 version. Tasks MSC-1-1 through MSC-1-7 cover the tasks required for a new installation and configuration of the software suite. All MARS members should review these tasks even if they have already updated the MSC software. The operational tasks, MSC-1-8 through MSC-1-11 assume configuration is according to tasks MSC-1-1 through MSC-1-7.

Instructions:

If not already completed, download MSC ver Feb 1 2021 from www.usarmymars.org. If not already installed and configured, complete tasks MSC-1-1 through MSC-1-7.

If MSC ver Feb 1 2021 has already been installed and configured, it is **not** necessary to re-install. Review the below configuration tasks against your configuration; note and correct any differences.

Mission Essential Tasks MSC-1-8 through MSC-1-11 are intended to eliminate mis-interpretations about how MSC is utilized and standardize operational concepts of the MSC across the MARS enterprise. Trainers, Mentors and other leaders should pay particular attention to the explanations in the notes and lead conversations that result in the adoption of these METs.

Figure 1 depicts the MSC software as it is installed on a single PC and is typical of most installations. More complex installations that support multiple radios and span locations across a local area network are in use by MARS members, but are beyond the scope of this training.

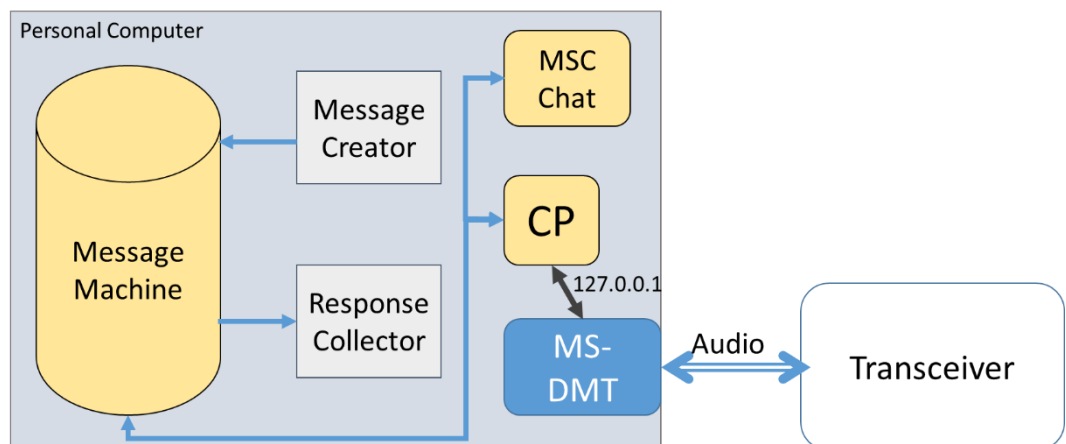


Figure 1 MSC Schematic Diagram

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Resources

The MSC installer media and installation instructions are available at:

<http://www.usarmymars.org/resources/software>

Go to MSC Suite, MSC Suite Installers and download

MSC Suite Install Instructions.pdf

And either:
MSCSuite-Install&Update-32bit-2021-Feb-01_1217.exe.wzd
or
MSCSuite-Install&Update-64bit-2021-Feb-01_1217.exe.wzd

If you are using a 64-bit operating system like Windows 10, the 64-bit version is recommended.



Additional

Notes:

In almost all cases, installing a MSC update over a previous version of MSC retains the previous configuration and ensures proper function of the new software with minimal effort.

Normally, backing up the previous version of MSC is **not** necessary.

In those cases where members choose to back-up the old MSC installation, the steps described here ensure the correct method that retains the previous configuration information is utilized.

The MSC Chat application is not listed as required, as later workflow training will minimize its use. Members are free to install MSC Chat if they wish.

Task Number:	Task Title:		
MSC-1-1	Install /Update MSC Software Suite		
Action: Install or Update MSC Software onto the PC.			
Condition: Given: - A personal computer and the ability to perform tasks in the Windows operating system. - Ability to use Encryption Wizard (EW) Software and in possession of the valid SDO passphrase. - Possession of MSC installer file from http://www.usarmymars.org/resources/software - Has read the MSC Software Suite Install Instructions.pdf In the case that a previous version of MSC is installed, it is assumed it functions correctly and is located in the default C:\MSC location.			
Standards: - Stores MSC installer media for later use. - Correctly install or update at a minimum the Communications Processor (CP), Message Creator (MC), Message Machine (MM) and Response Collector (RC) applications. - Demonstrates the correct method to back up a previous version of MSC if a backup copy is desired.			
Performance Steps Review MSC Software Suite Install Instructinons.pdf and watch the video How to Install MSC.Chuck Dillon.mp4 available at http://www.usarmymars.org/resources/software and perform the following steps: - Save a copy of the MSC installer file to your software archive. - If updating an existing MSC and a backup of the previous MSC is desired: locate existing C:\MSC folder and make a copy of the <u>entire MSC folder</u> as a backup. Be sure to use the copy function and <i>do not</i> simply rename this folder. - Run the installer program. Read and accept the license. Install to the default C:\MSC location. - Install at a minimum the following applications: CP, MC, MM and RC. - Validate the MSC software applications will execute from the shortcuts in the desktop folder. - Run each application. If this is an update to an existing MSC, validate that basic configuration information such as the station call signs are retained.			
Training and Evaluation: This task is self-evaluated by the Member.			
Performance Measures (PM):		GO	NO-GO
1. Member saved a copy to software archive		GO	NO-GO
2. Member backed up the previous MSC version using the Copy function		GO	NO-GO
3. Member selected and installed the required application(s) and validated their function.		GO	NO-GO
Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO; show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.			
References: Required - MSC Software Suite Install Instructions.pdf How to Install MSC.Chuck Dillon.mp4 video			
Warnings: None			
Notes:			



Resources

The MSC installer media, and MS-DMT modem are available at:
<http://www.usarmymars.org/resources/software>

The Communications Processor (CP) User Manual is accessed from the General tab of the configuration dialog. Open the configuration dialog by pressing the "gear" icon. Press the button labeled "User Manual" located near the left middle of the window.



Additional

Notes:

The Communications Processor (CP) is the interface between the modem and all applications. It is a server on the local host or LAN, and the applications connect to it in a server-client relationship.

Communications processing tasks such as encryption is performed in the CP.

Although the MSC can support hardware modems using a serial port, the configuration discussed here is between MSC and the MS-DMT modem using the MS-DMT IP interface.

Only the MS-DMT IP interface is recommended when using MS-DMT. All other methods to connect to MS-DMT are obsolete. This task only covers the steps to select the IP interface in MS-DMT.

Note: The Order-wire function in the Modem dialog window (M) always transmits without encryption, also known as Plain Text (PT). It is appropriate for troubleshooting and tests, not for communications purposes where encryption is required.

Task Number: MSC-1-2	Task Title: Configure Communications Processor			
Action: Configure the Communications Processor (CP)				
Condition: Given: - A personal computer and the ability to perform tasks in the Windows operating system. - Installed CP software (task MSC-1-1). - Has read the <u>Communications Processor User Manual.pdf</u>				
Standards: Configures CP with MS-DMT so that data is successfully transmitted and received through the data modem.				
Performance Steps Review <u>Communications Processor User Manual.pdf</u> and perform the following steps: - Configure the General tab. Enter your call sign, check use progress bar, and ensure “Use Review/Retry Queue” is unchecked. Check Minimize to System Tray and edit Position Identifier if more than one CP is used. See pp9 in User Manual. - Configure Cypher tab. See pp 10 in User Manual. - Configure Data Interfaces Tab. Set Device Type to MS-DMT TCP, check Show Modem Dialog on Start. Check Autostart Modem and set path for modem (for instance: C:\MSC\MS-DMT\MSDMT_64.EXE). Check Reset MS-DMT when data is 10% over payload size (to eliminate run on). Set RX Data Timeout for your system (varies based on modem used and your PC latency). Set Delay to Next TX to 5 Seconds (may vary based on modem used and PC latency). See pp 10-13 in User Manual. - Configure IP Interfaces tab. Check Show Only IPv4 Addresses. Set Server Listen Address, select either the local host address (if all applications run on the same machine), or the LAN segment. See pp15 in User Manual. - Set Up MS-DMT for IP. In configuration area, check TCP/IP. Set Data Port to 0. If more than one MS-DMT is utilized see specific instructions in MS-DMT help file. - Open the Modem dialog window by clicking the “M” icon on the CP main window. Type the word “test” in the Order-wire text line and press enter. Observe interactions between CP and MS-DMT in the Modem Status window and confirm proper function. (Order-wire text is PT)				
Training and Evaluation: This task is self-evaluated by the Member.				
Performance Measures (PM): - Member configured the General tab. CP operates minimized in System Tray and Position Identifier is a unique name for each CP being used. - Member configured Cypher tab. Allow PT is checked for QRPX and unchecked for regular MARS operations. - Member configured Data Interfaces tab for MS-DMT TCP. The Modem Status display in CP shows command and status exchanged between MS-DMT and CP. - Member configured IP Interfaces tab. Selected IPv4 and set server to either local host or LAN segment as appropriate for their situation. - Member set up MS-DMT for IP connectivity. The Modem Status display in CP shows command and status exchanged between MS-DMT and CP. - Member successfully sent test transmission from Modem Status window using MS-DMT.			GO	NO-GO
			GO	NO-GO
			GO	NO-GO
			GO	NO-GO
			GO	NO-GO
			GO	NO-GO
Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO; show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.				
References: Required Communications Processor User Manual				
Warnings: None				
Notes:				



Resources

The Message Creator (MC) User Manual is accessed from the dialog bar on the top of the MC window. Select User Manual in the Help drop down.

The [How to Update Address Book in MSC.Chuck Dillion.mp4](http://www.usarmymars.org/resources/software) video is available at: <http://www.usarmymars.org/resources/software>



Additional

Notes:

The Message Creator (MC) application is utilized to create formatted messages in MARS. It utilizes text-based templates and therefore can be updated and modified in the field to meet new requirements.

Task Number: MSC-1-3		Task Title: Configure Message Creator	
Action: Configure the Message Creator (MC) application			
Condition: Given: - A personal computer and the ability to perform tasks in the Windows operating system. - Installed MC software (task MSC-1-1). - Has read the Message Creator User Manual.pdf and viewed the How to Update Address Book in MSC.Chuck Dillion.mp4 video. - Has received the RGN_AB.csv file from Region Leadership.			
Standards: Configures MC so that formatted Message-Objects can be created using the appropriate call signs and plain language addresses; and Message-Objects are sent to the Message Machine as default action.			
Performance Steps Review Communications Processor User Manual.pdf , and How to Update Address Book in MSC.Chuck Dillion.mp4 video, Perform the following steps: - Copy the RGN_AB.csv file into the MSC/shared_data folder. Ensure you have AMARS_AB.csv and Global_AB.csv. Delete HQ_GAB.csv as it is obsolete. - Configure the Settings Tab, Station Information. Set ADDRESS_MYCALL= to your assigned call sign. Set ADDRESS_MYPLA= to your individual MARS Station PLA. Set ADDRESS_MYRI= to the RI of your State Tributary Station. Set MGRS_GRID. For multiple call signs, PLA, RI, etc list in comma separated without spaces. See pp13 in the User Manual. - Configure the Settings Tab, Templates. Check all 16L templates, Voice and Personal Contact Information. Uncheck all ACP templates (ACP is obsolete). See pp14 in the User Manual. - Configure the Personal Address book. Go to Addressing/Conditions tab. Next to Quick Add button select PERS_AB.csv. Press Quick Add. Enter all relevant information in the Quick Add dialog box, including Call-Sign, MARS Station PLA, and your RI (this is usually the RI of your State's Tributary Station. See pp 9 in the User Manual. - On bottom of all 16L tabs in the Processing group box. Check the box next to COMP. In Connections group box select Message Machine and press Default/Auto to ensure that the Messages Objects created in MC are never sent directly to the CP. See pp 11-12 in the User Manual.			
Training and Evaluation: This task is self-evaluated by the Member.			
Performance Measures (PM): - Member copied the RGN_AB.csv file. Region addresses are show in the Address/Conditions tab. - Member configured personal information on Settings Tab. Correct call signs and PLA are available in the MSGID line in templates. - Member uses only 16L, Voice and Personal Contact templates. - Member's information is available in the Address/Conditions tab. - Member has configured 16L tabs to send completed Message-Object to MM as default action.		GO	NO-GO
		GO	NO-GO
		GO	NO-GO
		GO	NO-GO
		GO	NO-GO
Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO; show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.			
References: Required - Message Creator User Manual How to Update Address Book in MSC.Chuck Dillion.mp4 video.			
Warnings: None			
Notes:			



Resources

The Response Collector (RC) Instruction Manual is accessed from the dialog bar on the top of the RC window. Select Instruction Manual in the Help drop down.



Additional

Notes:

The Response Collector (RC) application manages Data Products that are received in strip format.

The RC scans Message Objects and extracts the Data Strips that match the format programmed in the database. The strips are stored, sorted and may be recalled and consolidated.

Task Number: MSC-1-4	Task Title: Configure Response Collector		
Action: Configure the Response Collector (RC) application			
Condition: Given: • A personal computer and the ability to perform tasks in the Windows operating system. • Installed RC software (task MSC-1-1). • Is in possession of the specific Request Strip(s) in which data will be collected. • Has read the Response Collector Instruction Manual.			
Standards: • Configures RC on the correct LAN segment or local host. • Correctly enters valid Request Strips and establishes a database table for each strip.			
Performance Steps Review the RC Instruction Manual. Perform the following step for basic configuration. • Configure the Settings tab, Settings. Uncheck “Show Request Generator Tab.” Set IP Settings, select either the local host address or LAN segment. See pp11 in User Manual. Check “Only accept data sets that match respective input masks.” • Enter the Request Strip on the Form Input Tab. Paste the complete Request Strip into the Paste Request Strip box. Press “Generate Form.” Check on List or Database tab under Select Table drop down to validate a table for the Request Strip has been created. • Ignore Connections Dialog in lower left.			
Training and Evaluation: • This task is self-evaluated by the Member.			
Performance Measures (PM): 1. Set IP address so that RC is visible to applications as a server on either the local host or LAN. Validate that RC is running as a server by checking in MC or MM. 2. Enter Request Strip in Input Tab. Validate that a table for the Request Strip has been created.		GO	NO-GO
		GO	NO-GO
Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO; show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.			
References: Required • Response Collector User Manual			
Warnings: None			
Notes:			



Resources

The Message Machine (MM) Instruction Manual is accessed from the dialog bar on the top of the MM window. Select Instruction Manual in the Help drop down.



Additional Notes:

The Message Machine (MM) application stores and databases all Message Objects prior to transmission, or as they are received. The MM contains an InBox that is intended as the primary tool for reviewing and reading Message Objects.

MM performs robust logging functions. All Message-Objects should be routed through MM.

MM is built around a robust SQL database that allows stored content to be searched in a variety of ways, including from a remote terminal.

Rules can be created by the operator to automate certain functions in the MM. These rules allow MM to function as a "router" for Message Objects.

MARS Auxiliaries should take time to explore MM more advanced features after mastering the basic configuration in this Task.

Note: Processing a message is effectively removing the message from the active traffic list. It is in effect, archiving the message. Deleting or editing Message Objects is not allowed in MM so that custody of all events is retained for forensic investigation.

MM will only respond or acknowledge automatically if addressed specifically by its call sign. "ALL" should never be used as a MM call sign.

Task Number: MSC-1-5	Task Title: Configure Message Machine			
Action: Configure the Message Machine (MM) application				
Condition: Given: - A personal computer and the ability to perform tasks in the Windows operating system. - Installed MM software (task MSC-1-1). - Has read the Message Machine Instruction Manual.				
Standards: Configures MM with appropriate call signs and Routing Indicators that enable Machine Mode only when the MM is addressed specifically. Creates a Routing Rule to Response Collector.				
Performance Steps Review the Message Machine Instruction Manual. Perform the following step for basic configuration. - Configure the Settings tab, General. Enter call sign(s), ALE address, and RI that you are guarding. List only an RI that will be considered delivered and action is complete if received at your station. See pp9 in the User Manual. - Configure Settings tab, Network/Streams. Set IP Address for TCP server, select either the local host address or LAN segment. See pp10 in User Manual. - Set up Streams. Start all CP you will use. At bottom of screen, select each CP, check “ACK messages”, and press “Add/Update stream.” See pp 10-11 in User Manual. - Configure Settings tab, Machine Mode. Under Accept Messages, select “All Messages.” Under Machine Mode, select “Enable Machine Mode.” Uncheck “Disable Acknowledgement.” Set delay as appropriate for modem and PC latency. Check “Machine Mode Only on Encrypted Circuits,” check “Allow Multi-Message ZEE.” Set Message Size Limit based on transmitter duty cycle. See pp13,14 in User Manual. - Configure Settings tab, Routing. Start Response Collector. Check box “Enable Routing.” Under Trigger, type “*.” Under Action/Position ID select “Response Collector” from drop down. Leave everything else as is. Click “Add Rule.”				
Training and Evaluation: This task is self-evaluated by the Member.				
Performance Measures (PM): - Entered call sign(s), ALE address and RI in the proper format, and only RIs that are considered delivered to their final destination. No wildcards are utilized. - Set IP address so that MM is visible to applications as a server on either the local host or LAN. - Set up Stream to all CP in use. When MM and CP are running, a green banner indicated “Connected” is displayed in the Stream list. - Configured Machine Mode. Machine mode is enabled, delay is set to at least the default setting, and Machine Mode Only on Encrypted Circuits is checked. - Routing rule is established to Response Collector (RC). Messages that contain data strips match the RC’s database are displayed in RC when the rule is run.			GO	NO-GO
			GO	NO-GO
			GO	NO-GO
			GO	NO-GO
			GO	NO-GO
Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO; show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.				
References: Required Message Machine Instruction Manual				
Warnings: None				
Notes: The User Interface can be simplified using the Config menu bar drop down. The member may elect to hide unused tabs such as Sync and Reports. A configuration wizard is available in the Config menu bar drop down. The configuration wizard represents an alternate way to configure the MM.				



Resources

The MSC Chat User Manual is accessed from the dialog bar on the top of the MSC Chat window. Select User Manual in the Help drop down.



Additional

Notes:

The MSC Chat provides an on-air text-chat capability. Chat with all stations on the channel, or chat-room segregated modes are supported.

MSC Chat, nor any terminal application for that matter, should be used to cut/paste Message Object text. The MSC Chat configuration standard includes closing the multi-line text box so that pasting text into this window is not possible.

Task Number: MSC-1-6	Task Title: Configure MSC Chat	
Action: Configure the MSC Chat (CC) application		
Condition: Given: - A personal computer and the ability to perform tasks in the Windows operating system. - Installed MSC Chat (CC) software (task MSC-1-1). - Has read the CC User Manual.		
Standards: Configures CC with appropriate call signs for basic configuration. Tracks changes in MS-DMT in the CC modem dialog.		
Performance Steps Review the CC User Manual. Perform the following step for basic configuration. - Configure the Settings Tab. Enter call signs(s) in the Guarded Call Sign line. Uncheck “Ignore Messages Not For Me.” Check “Track Changes in CP.” - Configure Terminal Window. Move slider bar that separates received text box and multi-line box so that multi-line text area is eliminated and pasting of Message-Object text into multi-line area is not possible. Start CP. In connections group box, lower left corner, select the CP and press “Default/Auto.” In processing group box, lower left corner, check “Comp” and select “ZYG1”. Press “Set Default” button. - Set Modem to 2400/S. Type “Test” in Chat text box and press enter. Observe MS-DMT and ensure modem/interleave was set on MSDMT and the word “test” was transmitted. - Configure Machine tab. Enter the call sign of the message machine you will query in the “add machine address” text box. Do-not append with –CM. Hit “Add.” Verify entry by the machine address drop down.		
Training and Evaluation: This task is self-evaluated by the Member.		
Performance Measures (PM): - Configures CC with call sign(s). - Configures Terminal window so pasting Message-Object text is not possible - Sets default CP and encryption key. - Successfully completed test through to MS-DMT using the chat text bar. - Verified that remote message machine address was added in the Machine tab.	GO	NO-GO
	GO	NO-GO
	GO	NO-GO
	GO	NO-GO
	GO	NO-GO
Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO; show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.		
References: Required MSCChat Instruction Manual		
Warnings: None		
Notes: The User Interface can be simplified by hiding unused group boxes in the Terminal area. Select the Config UI drop down from the menu bar.		



Resources

The MSC installer media, and MS-DMT modem are available at:
<http://www.usarmymars.org/resources/software>



Additional

Notes:

The MSC suite was designed to be self-contained inside the MSC file structure. It does not utilize the Windows registry to store configuration data.

The entire MSC suite can be copied and run from any other location so long as the MSC folder structure remains intact. For instance, MSC can be run from a secondary hard drive, or removable media such as a thumb drive.

Task Number: MSC-1-7	Task Title: Create Back Up of MSC Suite	
Action: Create an active back up of the configured MSC Suite.		
Condition: Given: - A personal computer and the ability to perform tasks in the Windows operating system. - Installed MSC software (task MSC-1-1). - A USB thumb drive or other removable media.		
Standards: Successfully creates a recovery backup of the entire MSC folder on removable media.		
Performance Steps Perform the following step to create a recovery copy of MSC. - Insert removable media such as a USB thumb drive into the PC. Note the drive letter. Ensure the removable media is empty and has sufficient capacity to copy the entire MSC suite. - Locate the MSC folder. Right click on C:\MSC. Select Copy. - Locate the removable media. Right click and select Paste. NOTE: DO NOT DRAG AND DROP. This step may take some time to copy all the files. - Copy the MSC installer file to the removable media. - When complete, compare the files and folders in the C:\MSC with those copied to the removable media. Check the file and folder names exist in both places, and that the files are the same size. Finally, try running the various MSC programs from the back up drive. - Label and Store the recovery copy of MSC in a safe / secure place.		
Training and Evaluation: This task is self-evaluated by the Member.		
Performance Measures (PM): - Copies entire MSC folder using Copy/Paste technique - Validate that files are copied - Labels removable media to indicate it contains the MSC backup. - Stores removable media in appropriate location.	GO	NO-GO
	GO	NO-GO
	GO	NO-GO
	GO	NO-GO
Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO; show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.		
References: Required MSC Chat Instruction Manual		
Warnings: None		
Notes: Removable media provides a backup of the entire MSC suite and its configuration. The installer file should also be backed up. In the case of a catastrophic data loss on the PC, the MSC backup can be copied directly onto the computer to recover the suite. COPY/PASTE operation is preferred to Drag and Drop because the original files are always preserved.		



Resources

The Message Creator (MC) User Manual is accessed from the dialog bar on the top of the MC window. Select User Manual in the Help drop down.

The Message Machine (MM) Instruction Manual is accessed from the dialog bar on the top of the MM window. Select Instruction Manual in the Help drop down.



Additional

Notes:

Your station's address book entry should be at the top of the address book if you entered it according to the instructions in task MSC 1-3.

Message Creator is where Message-Objects are drafted. Once the Message-Object is created and sent to any other application it cannot be edited.

The "Digest" is a cryptographic hash that validates the data in the Message-Object is the same as what was drafted. Any attempt to edit the message, including "cut and paste" text will result in the Digest not being valid.

Message Machine should always be set as the default connection from Message Creator as all Message-Objects must be processed through Message Machine prior to transmission.

Message-Objects cannot be deleted in Message Machine. They are purged after one year. See settings in Database tab. Marking a message-object as Processed is effectively "archiving" the object and removes it from the active workspace. Processed messages can be viewed at any time by checking the box Incl. Processed or Show Processed on the "messages" tab.

Task Number: MSC-1-8	Task Title: 16L Message using MSC		
Action: Create, Transmit and Process a 16Line Message			
Condition: Given: • A personal computer and the ability to perform tasks in the Windows operating system. • Installed MSC software (task MSC-1-1). • Configured MM and MC software (task MSC-1-3, task MSC-1-4) • Transmitting and receiving equipment operating on authorized radio frequency/network			
Standards: • Successfully creates a 16L message in MC and processes it through MM.			
Performance Steps Perform the following steps: • Open Message Machine and Message Creator • In Message Creator, Data Input tab. Select any free form or USMTF template and enter test text. (MC User Manual pp7) • In Message Creator, Addressing/Conditions tab. Select the entry for your MARS station from the address book and add as the FROM (FM) label. Type your region number as the word spelled out (for example TEN) in the search bar. Select your Region's RJMOG from the search results and add it as the TO label. • In Message Creator, Addressing /Conditions tab. Set Conditions drop down to Precedence. Select Routine from the Value drop down and press Assign Condition. Set Conditions drop down to Classification, Value drop down to Unclassified and press Assign Condition. • In Message Creator, select "16L P2P abbr" tab. Review information and make any corrections in the Data Input or Addressing Conditions Tab. In the Connections group box select "Message Machine" then press "Send Text". • In Message Machine InBox. Find and validate the message you just sent to MM by clicking once and viewing in the right Message Info area. At bottom of screen, select "CP" as the connection, "ZYG1" as the encryption key and check the box next to Compress. Press "Send" to transmit the message. • Open the Message-Object by double clicking on it. Go to Msg Event Log. Note the date time it was received, transmitted and view are noted. Press "Mark as Processed" when you are ready to clear the message from your active workspace.			
Training and Evaluation: • This task is self-evaluated by the Member.			
Performance Measures (PM): 1. Correctly creates a Message-Object in the MC software. Enters all address and condition values required to activate the 16L P2P tab. 2. Reviews Message-Object in MC before sending and makes any corrections. Sends Message-Object to MM before transmission. 3. Locates Message-Object in MM. Opens Message Object and reviews Msg Event Log. 4. Select CP, encryption key, and compression. Transmits message.		GO	NO-GO
		GO	NO-GO
		GO	NO-GO
		GO	NO-GO
Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO; show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.			
References: Required • Message Creator User Manual • Message Machine Instructions Manual			
Warnings: None			
Notes:			



Resources

The Message Machine (MM) Instruction Manual is accessed from the dialog bar on the top of the MM window. Select Instruction Manual in the Help drop down.



Additional Notes:

This task highlights methods of monitoring data as it is being received in other than MSC chat.

If data is being received well and it includes a checksum header, a progress bar should pop up.

Fills are not allowed. Have the sending station re-send using more appropriate settings for the channel (change speed, interleave) or increase transmitter power. If RF communications is not possible with the station, then arrange for a relay through one of the other stations in the net.

Messages that contain errors that are accepted by MM will be placed into the yellow window on the bottom of the Message tab.

Checksum and Digest are both checked and noted in the Message Info tab in Green or Red text. If a message is received with bad checksum, there was an error in transmission. If a message was received with good checksum but failed digest, it indicates the sending station sent a corrupted copy.

All Message Machines that receive a QSL, either directly or by intercept, will note the call-signs and RI in the QSL and if the MM holds the message it will note the QSL action in its log for the message.

Task Number: MSC-1-9	Task Title: Receive and QSL a 16L Message										
Action: Receive and QSL a 16L message using MM											
Condition: Given: - A personal computer and the ability to perform tasks in the Windows operating system. - Installed MSC software (task MSC-1-1). - Configured MM and MC software (task MSC-1-3, task MSC-1-4) - Transmitting and receiving equipment operating on authorized radio frequency/network											
Standards: Successfully receives a 16L message in MM and sends a QSL.											
Performance Steps Perform the following steps: - Open Message Machine and Communications Processor. Check Settings, Network/Streams tab in MM and verify CP has a green “connected” banner. - Open the Modem Dialog window in CP by pressing the “M” button. Use this window to monitor the technical function of receiving data. Additionally, you may also monitor data on the LAN by switching to the Activity, In/Out tab in MM. - Arrange for another station to send a correctly formatted 16L message to you. Ensure good channel conditions that support at least good readable voice communications before sending /receiving data. Tell the other station to send and monitor the data as it is received. - Watch the CP Modem Dialog window while receiving. You should be able to see how the data is being received. If the data is heavily garbled and unrecognizable as a message object, do not expect it to be picked up by MM. If received successfully, it will show in your In-Box. - Send a QSL for the message from the MM In-Box. Click on the message(s) you wish to QSL and at the bottom of the In-Box screen select the encryption key and the CP. Press the “QSL” button. Observe the QSL message being sent in the CP Modem (M) window.											
Training and Evaluation: This task is self-evaluated by the Member.											
Performance Measures (PM): - Monitors incoming and outgoing activity using the Modem Dialog window in CP. - Receives good copy of a message after ensuring a good RF communications circuit that is likely to support data transmission. - After message is received, checks Checksum and Digest. - Sends QSL to the sending station.			<table><tr><td>GO</td><td>NO-GO</td></tr><tr><td>GO</td><td>NO-GO</td></tr><tr><td>GO</td><td>NO-GO</td></tr><tr><td>GO</td><td>NO-GO</td></tr></table>	GO	NO-GO	GO	NO-GO	GO	NO-GO	GO	NO-GO
GO	NO-GO										
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GO	NO-GO										
Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO; show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.											
References: Required Message Machine Instructions Manual											
Warnings: None											
Notes:											



Resources

The Response Collector (RC) Instruction Manual is accessed from the dialog bar on the top of the RC window. Select Instruction Manual in the Help drop down.



Additional Notes:

A Request Strip is an XML like strip that requests specific information in a specific format. It is used to create the data entry dialog in RC. A Request Strip is usually sent in a Request For Information (RI) message, which contains supporting instructions.

A Response Strip is the answer to a Request Strip. The Response Strip contains the data in the specific format required.

Request Strips and Response Strips combined create and populate a database with specific information. It is the way MARS is asked a question that requires a response composed of a large volume of data from many locations.

Task Number: MSC-1-10	Task Title: Create Response Strips			
Action: Create a Response Strip				
Condition: Given: - A personal computer and the ability to perform tasks in the Windows operating system. - Installed MSC software (task MSC-1-1) - Configured RC, MM and MC software (tasks MSC-1-3, MSC-1-4, task MSC-1-5) - Transmitting and receiving equipment operating on authorized radio frequency/network				
Standards: Create Response collector database with request strip and perform data entry tasks.				
Performance Steps Perform the following steps: - Open CP, MM, MC, and RC. - Receive an RI message that contains a Request Strip. Identify the Request Strip and the supporting instruction text. Highlight the Request Strip, right click and select copy. - Go to the Input, Form Input tab of RC. Right click and paste into the text box that is labeled "Paste Request Strip Here". Then press the "Generate Form" button. The Response Data Entry area will populate with labels, drop downs, or text lines. Note the "Select Existing Request" drop down will now contain the name of the Request Strip so it can be recalled for later use. - Complete the Date Entry area. Refer to the instructions in the RI and note the valid data notations in the text boxes. Enter only valid data. Press "Generate Response Strip" after data entry is complete. - Note the Response Strip is printed in the text box near the bottom of the window. Press the button "Copy Response Strip." - Go to MC, Data Input tab. Select "USMTF RRI Paste Response Strip.txt" template. Paste the Response Strip into the area of the template labeled "Paste RRI Strips". - Set "Addressing/Conditions" and select the appropriate 16L tab. Send the Message Object to MM. Select the Message Object from the In-Box and transmit to another station.				
Training and Evaluation: This task is self-evaluated by the Member.				
Performance Measures (PM): - Correctly identify a Request Strip and supporting instructions in a Request for Information Message - Creates a data entry dialog in Response Collector by pasting the Request Strip into the correct text box. - Correctly enters data, noting the valid data entry characters and the instructional text in the RI. Selects correct RRI template and pastes Response Strip into template. - Completes RRI template, 16L format, and transmits from MM.			GO	NO-GO
			GO	NO-GO
			GO	NO-GO
			GO	NO-GO
Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO, show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.				
References: Required Response Collector Instructions Manual				
Warnings: None				
Notes: Multiple Response Strips of the same type can be pasted into a single USMTF RRI message. It is generally considered poor form to put Response Strips of different types into the same RRI as this information may be going to different destinations. While RC will sort this information into different database tables, do not assume all destinations are in possession of the RC software.				



Resources

The Response Collector (RC) Instruction Manual is accessed from the dialog bar on the top of the RC window. Select Instruction Manual in the Help drop down.



Additional

Notes:

Forwarding Rules in MM allow messages that match certain criteria to be either retransmitted or forwarded on the local MSC system. This task depends on a forwarding rule that sends all Message-Objects to the Response Collector. This technique is preferred to simply connecting RC to CP because it allows the rule to be re-run on MM and supports data recovery from times when RC is not operating.

Army MARS has no standing rule for segmenting long messages. Rather we rely on the best judgment of experienced radio operators to determine how long a data packet can successfully pass in a single transmission, based on the existing circuit conditions.

Messages are consolidated in each Region. In some cases, they are first consolidated in the Tributary Relay Stations/Message centers before being passed up to the Region Minor Relay/Message Center. Consolidated messages are always sent from the Region as a finished product – either to the TransGlobal network or the EARS ALE relay stations.

Task Number: MSC-1-11	Task Title: Consolidate/Collect Response Strips										
Action: Use RC software to collect and consolidate Response Strips											
Condition: Given: • A personal computer and the ability to perform tasks in the Windows operating system. • Installed MSC software (task MSC-1-1) • Configured RC, MM and MC software (task MSC-1-3, MSC-1-4, task MSC-1-5) • Transmitting and receiving equipment operating on authorized radio frequency/network											
Standards: • Receives multiple Response Strips and processes these into the RC database tables for transmission in a single RRI message.											
Performance Steps Perform the following steps: • Open CP, MM, MC, and RC. • Receive a RI message that contains a Request Strip(s). Identify the Request Strip(s) and the supporting instruction text. Highlight and copy the Request Strip(s) into the RC according to task MSC-1-10 and note database tables are created for each one by checking the List or Database tabs. • Verify the forwarding rule described in task MSC-1-5 is in place. Re-run the rule by entering the rule number in the text box on the bottom of the MM Routing Tab and pressing the “Re-Run Rules” button. Note the database tables in RC will begin populating with data from the Message Machine. This process may take several minutes if the MM contains a lot of messages. • In the RC’s Database tab. Select the database table of each response type you are processing. Press the button “Remove Duplicate From Selected Table” button. Review each record in the table and perform quality control. Select and delete records that are in error. Follow up and question any records that contain suspected false or sensationalized information. • In the RC’s List tab. Select the database table of the response type you wish to consolidate and report using the dropdown labeled “Select Table”. Press the button “Copy Responses.” Paste response into the USMTF RRI Paste Response Strips.txt template according to the procedure in task MSC 1-10. Monitor number of strips in message and segment into multiple messages if appropriate (based on best judgment considering channel conditions on the network and ability to transmit/receive long messages). Transmit message.											
Training and Evaluation: • This task is self-evaluated by the Member.											
Performance Measures (PM): 1. Create Database Tables in RC for each Request Strip. 2. Receive RRI messages that contain Response Strips. Use forwarding rules to route these messages from MM to RC. 3. Remove duplicate entries from RC tables. Perform Quality Control on database by deleting entries in error and questioning suspected false information. 4. Copy entire database table to a consolidated RRI message(s). Demonstrate good judgment in segmenting into multiple messages. Transmit messages(s).			<table><tr><td>GO</td><td>NO-GO</td></tr><tr><td>GO</td><td>NO-GO</td></tr><tr><td>GO</td><td>NO-GO</td></tr><tr><td>GO</td><td>NO-GO</td></tr></table>	GO	NO-GO	GO	NO-GO	GO	NO-GO	GO	NO-GO
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Evaluation Guidance: Members passing all PMs receive a GO. Members not completing any PM receive a NO-GO; show what they did wrong and how to do it correctly. Have the Member perform the PMs until they perform the PMs correctly.											
References: Required • Response Collector Instructions Manual											
Warnings: None											
Notes:											